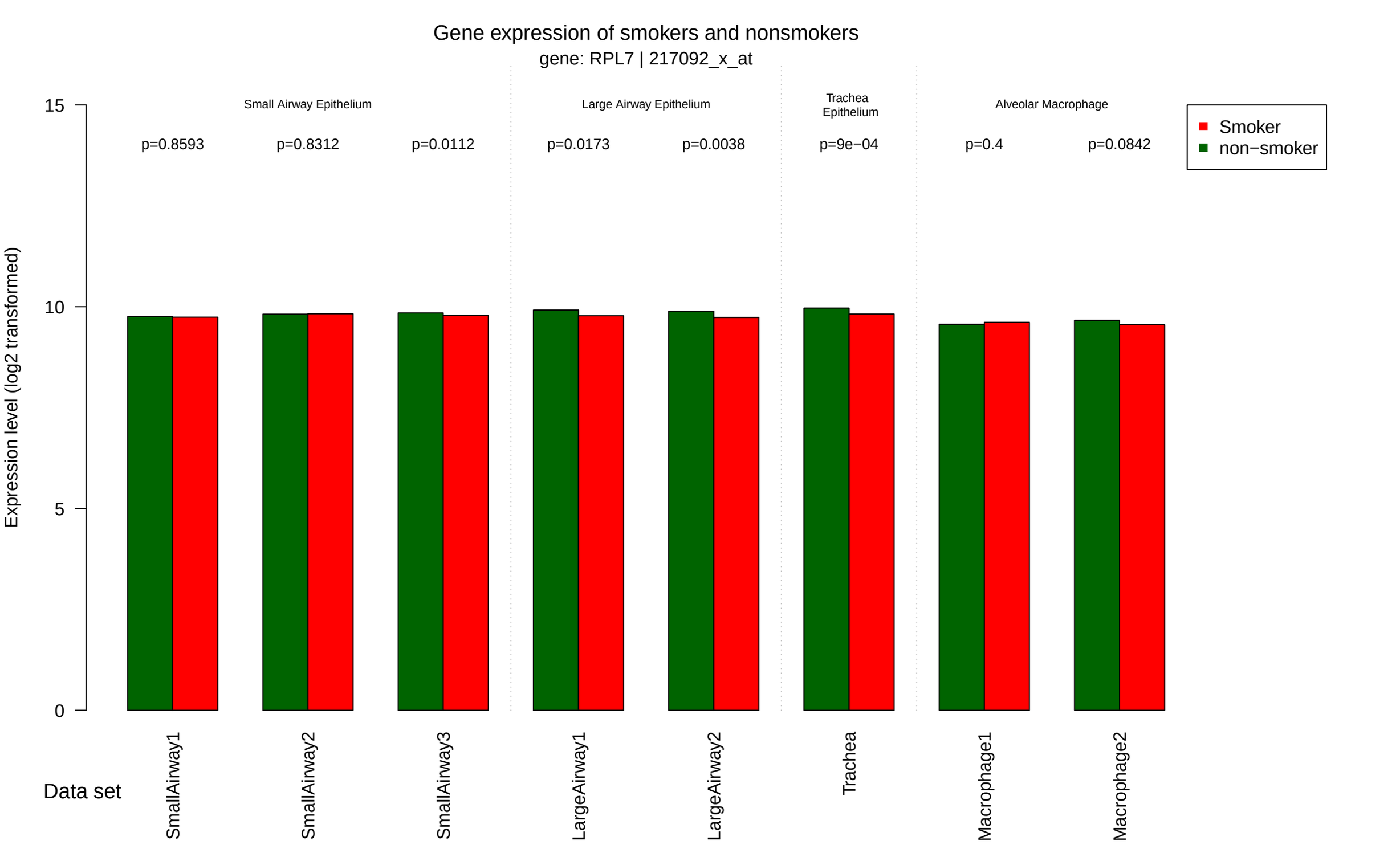
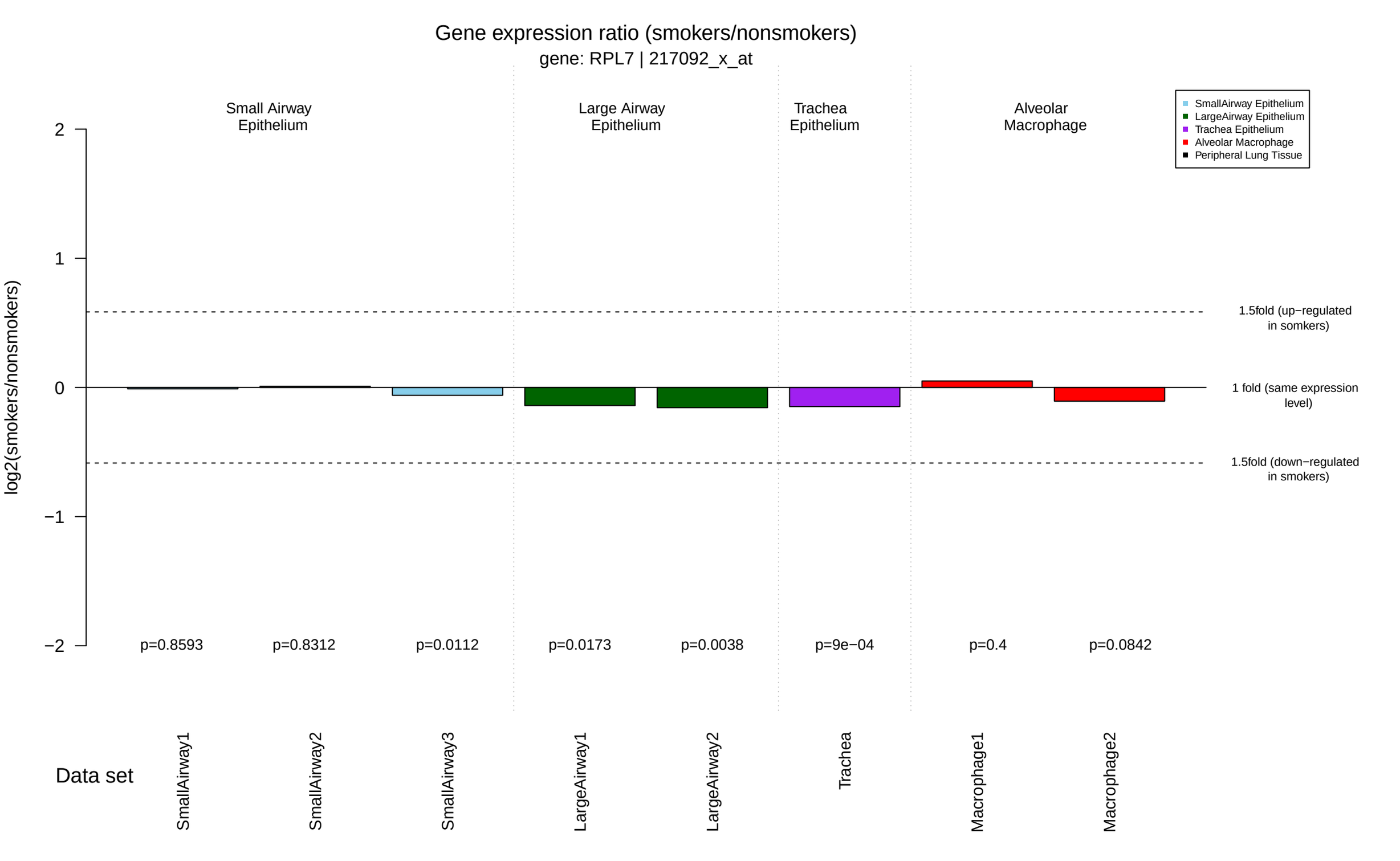
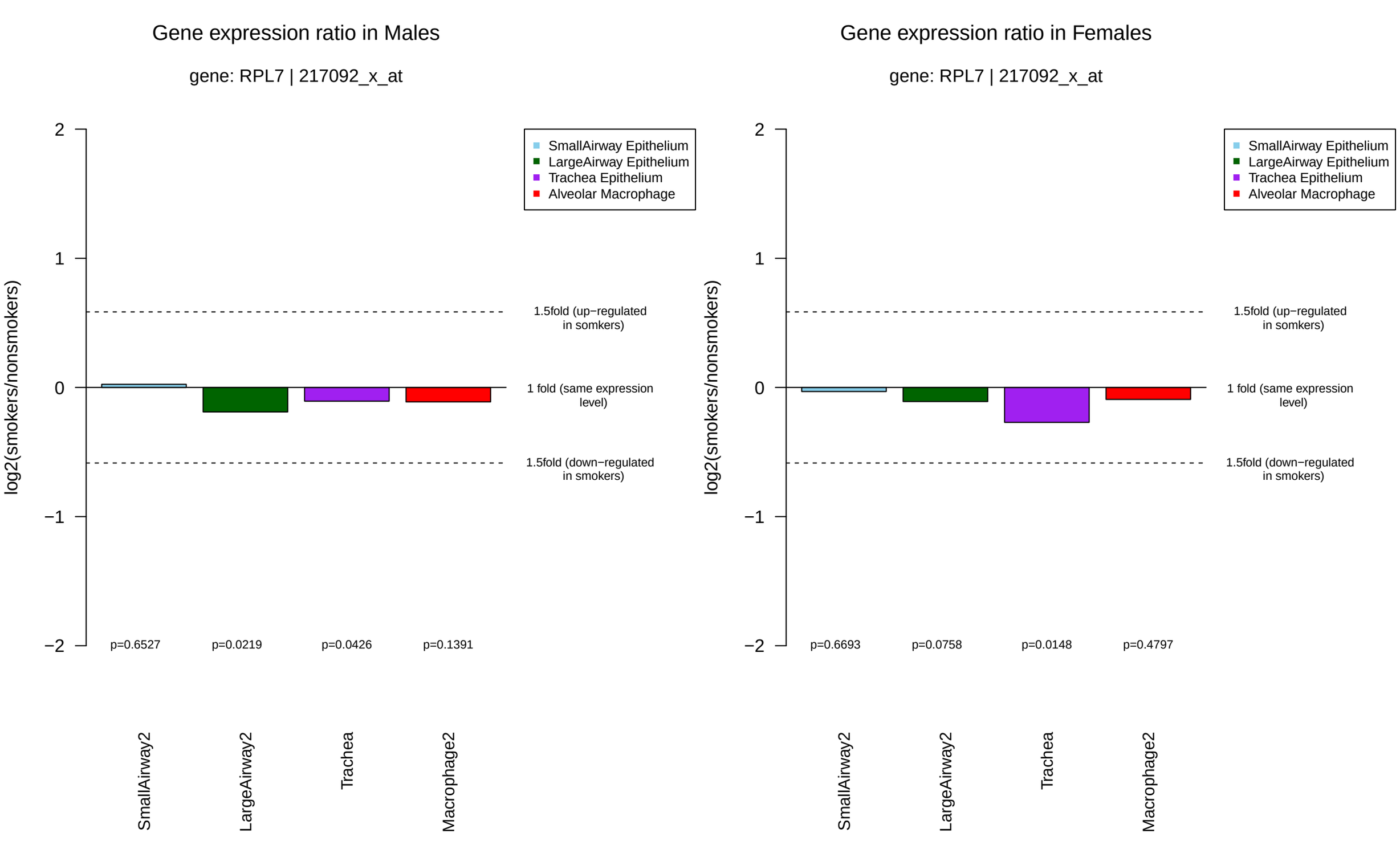
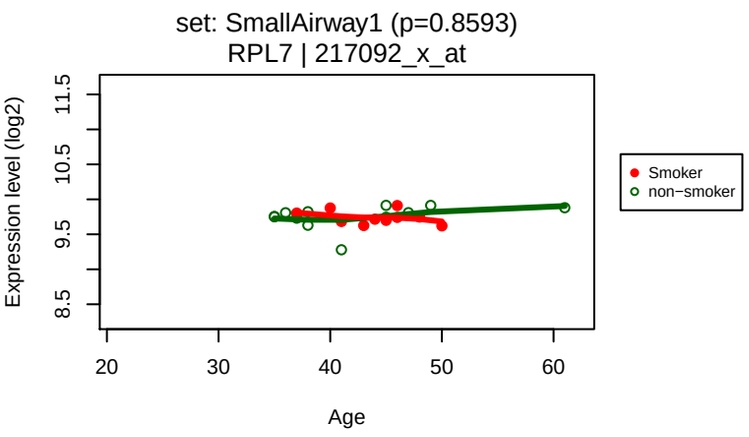
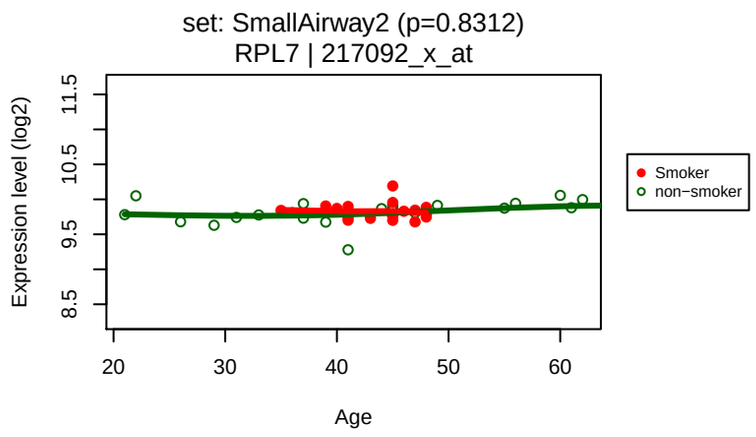
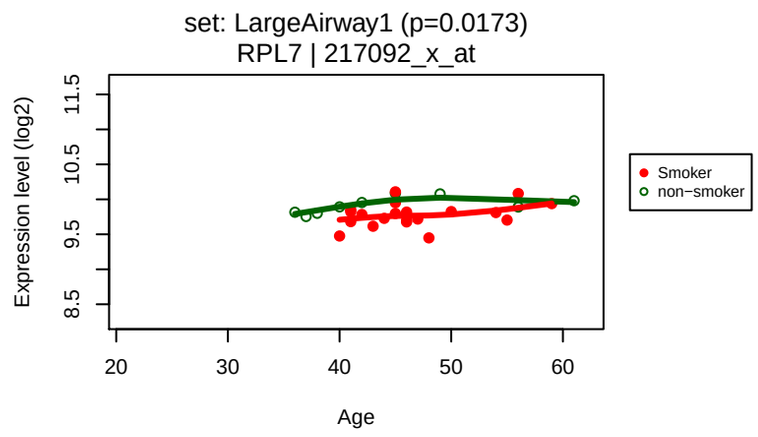

Gene expression of smokers and nonsmokers
gene: RPL7 | 217092_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8593
p=0.8312
p=0.0112
p=0.0173
p=0.0038
p=9e−04
p=0.4
p=0.0842
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

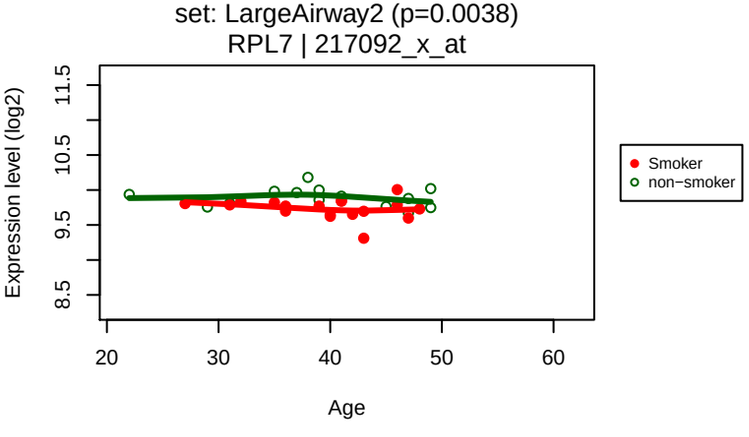

Gene expression ratio (smokers/nonsmokers)
gene: RPL7 | 217092_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8593
p=0.8312
p=0.0112
p=0.0173
p=0.0038
p=9e−04
p=0.4
p=0.0842
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

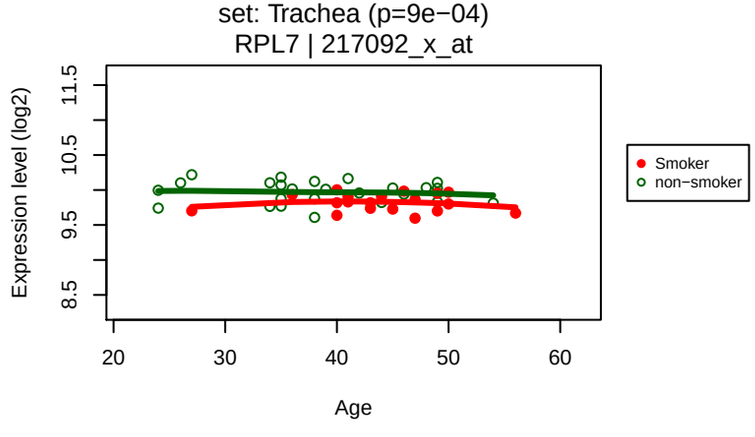

Gene expression ratio in Males
gene: RPL7 | 217092_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6527
p=0.0219
p=0.0426
p=0.1391
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL7 | 217092_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6693
p=0.0758
p=0.0148
p=0.4797
SmallAirway2
LargeAirway2
Trachea
Macrophage2

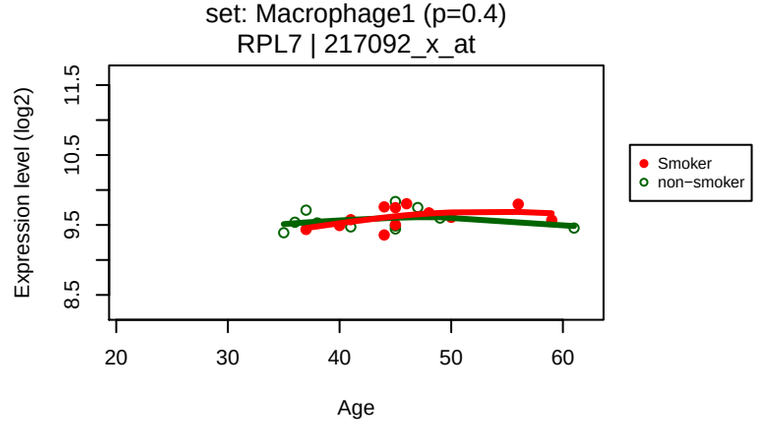

set: SmallAirway1 (p=0.8593)
RPL7 | 217092_x_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker

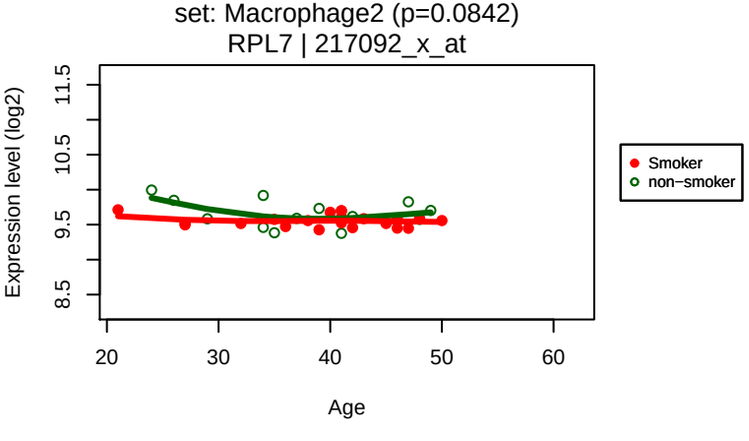

set: SmallAirway2 (p=0.8312)
RPL7 | 217092_x_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker

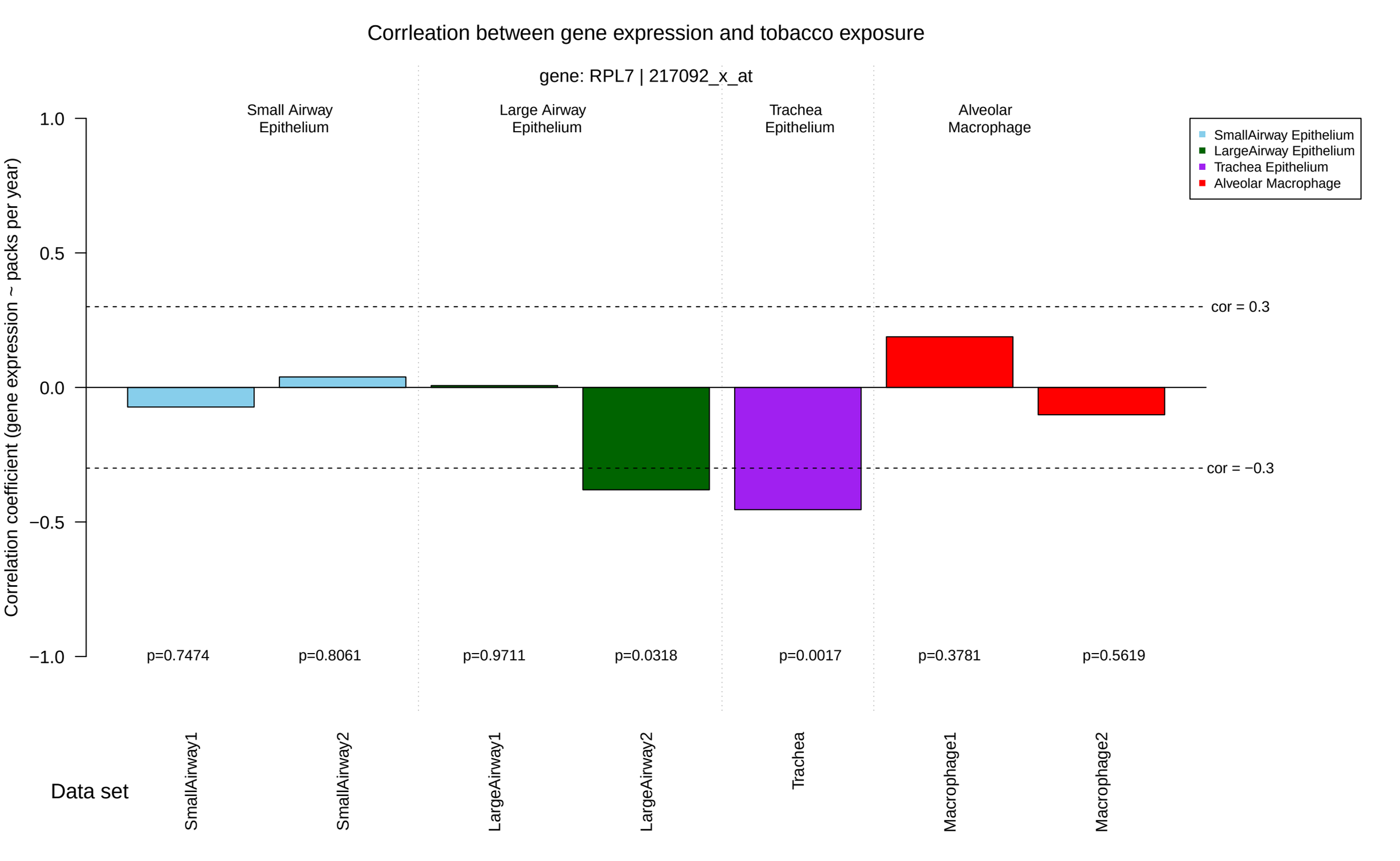

set: LargeAirway1 (p=0.0173)
RPL7 | 217092_x_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0038)
RPL7 | 217092_x_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=9e−04)
RPL7 | 217092_x_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4)
RPL7 | 217092_x_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0842)
RPL7 | 217092_x_at
Expression level (log2)
8.5
9.5
10.5
11.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL7 | 217092_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7474
p=0.8061
p=0.9711
p=0.0318
p=0.0017
p=0.3781
p=0.5619
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2